

Appendix E: Mathematical Derivation of BMI Resonance Modes

The empirical data recorded in Chapter 12—specifically the observed resonant peaks at 115.03 Hz (H1) and 130.03 Hz (L1)—necessitates a formal mathematical derivation of the BMI interface resonance modes.

1. Manifold Perturbation and Boundary Trapping

We model the manifold interface as a constrained boundary condition within a higher-dimensional bulk. The metric perturbation $h_{\mu\nu}$ near the interface satisfies the generalized wave equation:

$$(\square_g + M^2) h_{\mu\nu} = S_{\mu\nu}$$

where $\square_g$ is the d'Alembertian operator in 3+1 spacetime and M represents the mass-energy effective coupling constant of the interface. In the limit of a compactified hyperdimensional cavity of length L , the resonance modes ω_n are quantized according to the boundary topology:

$$\omega_n = (n\pi c / L) \times \sqrt{1 - \beta^2}$$

2. Derivation of the Frequency Split (Δf)

The observed frequency split $\Delta f \approx 15$ Hz is a consequence of the projection of the tensor field h_{ij} onto the detector orientation tensors. The frequency shift δf is determined by the geodesic deviation equation:

$$\delta f = (1 / 2\pi) \times \int_{\Gamma} \nabla_{\mu} (D_{ij} h^{ij}) dt$$

The ratio of the peaks $f_L / f_H \approx 1.13$ implies a non-linear curvature interaction coefficient $K \approx 0.13$, anchoring the physical dimensions of the interface nodes to a scale of approximately 10^{-15} m.